

ARISTOTLE UNIVERSITY OF THESSALONIKI

FACULTY OF ENGINEERING – SCHOOL OF MECHANICAL ENGINEERING

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SEMESTER: 8th

COURSE: DATA ANALYSIS

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1st ASSIGNMENT

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Note: figures are referenced by their original labels (e.g. “Image 2.1”). Replace each `\figplaceholder` with your own generated figure via `\includegraphics`.

1. Introduction

The purpose of this assignment is to define the dataset on which this and the remaining assignments will be based, and to separate the training data of the model that will be created from the data that will be used to test the model (test data). The data are taken from the file `Data_Analysis_2026.csv`. Then, the questions of the assignment are answered in order. Each answer presents the methodology that was applied, the portion of code that was written, the results it produced, and finally a commentary. Finally, the model will continue to be developed over time, as it will be used in the following assignments. The problem that was given examines data concerning wine production and contains information about fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol, quality, and wine type.

2. Preprocessing

The first and most fundamental actions are defining the data we will work on from the initial set contained in the file. This is done through the `sample` command contained in the **pandas** library, which was called in the first lines of the code together with **numpy**. The seed with which the `sample` function operates is the student ID, AEM = 7137. The training data are stored in `sarxeio` (`sample_arxeio`) and contain 80% of the total data, while the rest (20%) is automatically assigned to the test data in the variable `tarxeio` (`test_arxeio`). In addition, with the commands `sarxeio.info`, `sarxeio.describe` and `sarxeio.head()` we obtain some initial information about the data, such as how many original records are in the file (rows), what kind of data each column contains (str, float64), and how many rows contain not-null entries (i.e. at least one character is written).

[Image 2.1 — insert your figure here: `sarxeio.info()` output]

Image 2.1

3. Question 1

In the first question we are asked to identify errors, invalid values, and duplicate records. First, we start from the simplest task, which is removing the duplicate records, i.e. rows that were recorded more than once. To begin, we assign to the variable `double` the task of finding and counting the duplicates. It then displays the number of rows (6708) and the number of duplicates before cleaning (1115). With the `drop_duplicates` command, the variable `sarxeio` is “cleaned” of the duplicates. Finally, we confirm that no duplicate remains (0) and we see the number of remaining rows (5593).

[Image 3.1 — insert your figure here: rows before/after cleaning, duplicate counts]

Image 3.1

Once it was cleaned of duplicates, an analysis of the unique values for each column is performed. In other words, each column is checked separately, and every different value found while reading is stored

and displayed in a list as a string ('str', text). To do this we use the `for` loop available in Python. In this way we can manually check the unique values. In addition, the total number of different values contained in the list is given for greater convenience, along with the type of data contained ('str' / 'float64'). All rows contain strings except the last two, which contain almost no data in their 5593 rows, and this is the reason they will be deleted later in the code, since they do not contain any important information (they do not even have a name, Unnamed).

[Image 3.2 — unique values of the ALCOHOL column]

[Image 3.3 — unique value of the UNNAMED: 13 column]

Image 3.2 Image 3.3

In the given data, the last category, **wine type**, after reading the whole column, appears to be dominated by two categories: the white (white) and the red (red) wines. It is obvious that these two types of wine have slightly different properties, which are described in the remaining columns. Therefore, after the erroneous entries are normalized (due to typos), we split our original variable `sarxeio` into two subsets of it, `red_wines` (1380) and `white_wines` (3931), so that the data can be managed better. In addition, a further separation is made into clean values and invalid values, in which all the remaining types that could not be registered as red or white (−999, NaN, 999, 9999, ?) were placed and counted. These were not assigned to any variable and will be removed along the way.

[Image 3.4 — clean values (white / red counts)]

[Image 3.5 — invalid values found]

Image 3.4 Image 3.5

Next, the remaining columns are cleaned as well. Because all the remaining variables contain numerical (float) data, they go through a check, and in case a float is not found, the value is converted to NaN (Not a Number). The empty cells are then computed and stored as a sum in a variable `missing_values`. Finally, it passes through an `if` check to display the appropriate message and the value of the missing entries.

[Image 3.6 — missing values per column / cleaning completed]

Image 3.6

Then follows the “cleaning” loop. In this loop, the columns that have no data are removed and the useful columns are counted. This number will be used to create a “threshold” that keeps the rows having up to 3 NaN, since the remaining information is sufficient and should not be lost. In the cases where they are kept, the NaN values are replaced with the mean value of the column for the wine type in which they were registered. Beforehand, there is a command to compute the mean and the median for each variable per wine type. Then it is re-examined whether all empty cells have been filled, so that the process can continue.

[Image 3.7 — mean and median per wine type]

[Image 3.8 — check for empty cells after filling]

Image 3.7 Image 3.8

Question 2

In the second question we are asked to construct and comment on boxplot-type charts for all the continuous variables, first regardless of wine type and then separately the white from the red ones.

Because the charts are quite many, they are presented in the appendix, and the comments are placed in this paragraph.

Something that is immediately perceived is the fact that when charts are created for the whole dataset without taking into account the category to which they belong (white/red), the outlier values are more numerous. This occurs in each of the first 12 charts (Images 6.1–6.12). This is reasonable, since there are more values, the mean is between the two means, and a value that would have been within the limits ends up outside, because the lower limits have risen while the upper ones have fallen.

The fixed acidity appears not to be “fixed”, since in all three charts there are several values above the upper limit.

The extreme values of the volatile acidity (Image 6.2) appear to originate from the measurements of the white wines (Image 6.26), since there greater variability and more extreme values are observed compared to the corresponding chart for the reds (Image 6.14).

The same happens in the case of citric acid (Image 6.3), since in the red wines (Image 6.15) there appears to be only one extreme value, and that is at the upper limit. On the contrary, even in the chart of the whites (Image 6.27) there are several values on either side of the limits.

Conversely, in the case of residual sugars, most outliers appear to originate from the red wines (Image 6.4, Image 6.16, 6.28), since in the whites they appear to be contained within the 25–75% limits.

For the chlorides, the data appear relatively “gathered”, and in all three charts (Image 6.5, 6.17, 6.29) the extreme values are close to the limits; but in all three charts 2 values appear (one for white, one for red) that differ by an order of magnitude. We understand that this entry is erroneous and should be deleted or transformed.

The density of the red wines has large variability and has outliers both toward the upper and the lower limits. On the other hand, the whites have extreme values only at the upper limit, but they are not many in number.

The pH, since we know its physical meaning, makes it apparent whether there are erroneous entries, because in both types there is an entry with pH above 14 (excessively basic), whereas we know that wines are moderately acidic. There are also entries near or below zero; again this appears to be erroneous, but there could have been some error in production that produced these values (?).

The same happens for the alcohol charts, since the content is a percentage and there appears to be a measurement above 100% and near/below zero. These values can be deleted or normalized.

For quality, it appears that there were erroneous entries, or that 2 systems were used by mistake (0–10 and 0–100).

For the sulphates, it appears that most extreme values originate from the reds, without however implying any correlation, since both types have quite a few outliers.

For the remaining charts, there does not appear to be any noteworthy correlation.

4. Question 3

In this question we are asked to identify the extreme values (outliers) with the **IQR** method and with the **Z-score** method using a limit of 3.5 standard deviations.

Starting with the IQR method, we need to define the variable `numeric_cols_red`, which from `red_wine` selects the columns that contain only numerical data, so that the code runs only on those. Then we create a `for` loop in which, for each column separately, its data are checked. It creates two variables, `Q1` (which defines the lower 25% limit) and `Q3` (which defines the upper 75% limit). The $IQR = Q3 - Q1$, and to find the limits we work as follows:

$$\text{Lower limit (Lower)} = Q1 - 1.5 \times IQR$$

$$\text{Upper limit (Upper)} = Q3 + 1.5 \times IQR$$

The values are then checked to see whether they are lower than the lower limit or higher than the upper limit, and they are stored so that their total can be presented separately. The process is repeated for the white wines as well, the only change being the creation of `numeric_cols_white`, which checks `white_wine`.

For the z-score method, we want to find the values that are more than 3.5 standard deviations away from the mean value of the data. Any value that lies outside these limits (either $\mu + 3.5\sigma < \text{value}$, or $\mu - 3.5\sigma > \text{value}$) is considered an outlier. Thus, another `for` loop is created for each column. Inside it, the mean value (μ) and the standard deviation (σ) are computed. Then, the z-score is computed through the formula

$$z = \frac{x - \mu}{\sigma},$$

and it is checked with an operator whether the absolute value of z (`abs(z_score)`) is greater than 3.5. In the case where the hypothesis is `True`, then it is considered an outlier and is noted. Finally, a message is displayed that visualizes the results for each column for each method.

For the transformation of the extreme values, the **Clamp Transformation** method is used. Using the IQR method I create the upper and lower limits, and with the command `.clip(lower, upper)`, the values that lie outside the limits take the value of the limit.

[Image 5.1 — outliers (IQR) for RED WINES]

[Image 5.2 — outliers (Z-score) for RED WINES]

Image 5.1 Image 5.2

[Image 5.3 — outliers (IQR) for WHITE WINES]

[Image 5.4 — outliers (Z-score) for WHITE WINES]

Image 5.3 Image 5.4

5. Question 4

In the fourth question, certain characteristics are requested both for the variables and for the final dataset, which has now been “cleaned”. For the continuous and for the categorical variables, the requested characteristics are given.

[Image 5.1 — continuous variables statistics]

Image 5.1

[Image 5.2 — categorical variables statistics]

Image 5.2

And the relevant histograms, as well as a heatmap table (correlation matrix), are shown below.

[Insert your figures here: histograms (volatile acidity, quality, chlorides, residual sugar, sulphates, total sulfur dioxide, density, pH, alcohol, citric acid)]

[Insert your figures here: Correlation Matrix heatmap & Wine Type Distribution bar chart]

Appendix

Wine Figures

The boxplots referenced throughout the report are collected here. The labels below match the original numbering (Images 6.1–6.36): Images 6.1–6.12 are for all wines combined, Images 6.13–6.24 are for the red wines, and Images 6.25–6.36 are for the white wines.

[Image 6.1 — boxplot: fixed acidity (all wines)]

[Image 6.2 — boxplot: volatile acidity (all wines)]

Image 6.1 Image 6.2

[Image 6.3 — boxplot: citric acid (all wines)]

[Image 6.4 — boxplot: residual sugar (all wines)]

Image 6.3 Image 6.4

[Image 6.5 — boxplot: chlorides (all wines)]

[Image 6.6 — boxplot: free sulfur dioxide (all wines)]

Image 6.5 Image 6.6

[Image 6.7 — boxplot: total sulfur dioxide (all wines)]

[Image 6.8 — boxplot: density (all wines)]

Image 6.7 Image 6.8

[Image 6.9 — boxplot: pH (all wines)]

[Image 6.10 — boxplot: alcohol (all wines)]

Image 6.9 Image 6.10

[Image 6.11 — boxplot: quality (all wines)]

[Image 6.12 — boxplot: sulphates (all wines)]

Image 6.11 Image 6.12

[Image 6.13 — boxplot: fixed acidity (Red Wines)]

[Image 6.14 — boxplot: volatile acidity (Red Wines)]

Image 6.13 Image 6.14

[Image 6.15 — boxplot: citric acid (Red Wines)]

[Image 6.16 — boxplot: residual sugar (Red Wines)]

Image 6.15 Image 6.16

[Image 6.17 — boxplot: chlorides (Red Wines)]

[Image 6.18 — boxplot: free sulfur dioxide (Red Wines)]

Image 6.17 Image 6.18

[Image 6.19 — boxplot: total sulfur dioxide (Red Wines)]

[Image 6.20 — boxplot: density (Red Wines)]

Image 6.19 Image 6.20

[Image 6.21 — boxplot: pH (Red Wines)]

[Image 6.22 — boxplot: alcohol (Red Wines)]

Image 6.21 Image 6.22

[Image 6.23 — boxplot: quality (Red Wines)]

[Image 6.24 — boxplot: sulphates (Red Wines)]

Image 6.23 Image 6.24

[Image 6.25 — boxplot: fixed acidity (White Wines)]

[Image 6.26 — boxplot: volatile acidity (White Wines)]

Image 6.25 Image 6.26

[Image 6.27 — boxplot: citric acid (White Wines)]

[Image 6.28 — boxplot: residual sugar (White Wines)]

Image 6.27 Image 6.28

[Image 6.29 — boxplot: chlorides (White Wines)]

[Image 6.30 — boxplot: free sulfur dioxide (White Wines)]

Image 6.29 Image 6.30

[Image 6.31 — boxplot: total sulfur dioxide (White Wines)]

[Image 6.32 — boxplot: density (White Wines)]

Image 6.31 Image 6.32

[Image 6.33 — boxplot: pH (White Wines)]

[Image 6.34 — boxplot: alcohol (White Wines)]

Image 6.33 Image 6.34

[Image 6.35 — boxplot: quality (White Wines)]

[Image 6.36 — boxplot: sulphates (White Wines)]

Image 6.35 Image 6.36